

NIKHIL CHOWDARY TOKALA

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PROFESSIONAL SUMMARY

Strategic Data Science Manager & ML Systems Lead with 10+ years of experience designing high-throughput distributed ML systems, counterfactual inference frameworks, and Agentic AI solutions. Expertise in leading engineering teams, optimizing large-scale Spark compute pipelines, and driving core product growth across fast-paced technology environments.

SKILLS

Machine Learning & AI: Agentic AI Frameworks, RAG, Counterfactual Inference, Causal Inference, Uplift Modeling, XGBoost, Decision Trees

Experimentation & Analytics: A/B Testing, Time Series Forecasting, Predictive Analytics

Deep Learning & NLP: TensorFlow, Keras, PyTorch, Word Embeddings, LLMs, Computer Vision

Big Data & Cloud: PySpark, SparkSQL, AWS (Bedrock, SageMaker, EMR, S3, Glue), Airflow, Elasticsearch

Programming & Tools: Python, SQL, R, Linux/Shell Scripting

Engineering & DevOps: Git, Docker, CI/CD, Code Review, Mentorship

Data Visualization: Tableau, Power BI, Data Studio

GenAI: AWS Bedrock, Prompt Engineering, LLM Evaluation, Diffusion Models

EDUCATION

Postgraduate Program in Artificial Intelligence & Machine Learning *(Part-time)*
The University of Texas at Austin **GPA 4.0/4.0**

2023 - 2024

Bachelor of Technology, Aerospace Engineering
Indian Institute of Technology Bombay (IIT Bombay), Mumbai, India

2012 - 2016

WORK EXPERIENCE

TIGER ANALYTICS, Plano, Texas

Oct 2022 - Present

MANAGER, DATA SCIENCE

- Designed distributed ML systems and counterfactual inferencing models using PySpark to process multi-terabyte interaction datasets, evaluating unseen cohort performance to improve automated decision rules.
- Engineered an Agentic AI data analysis tool leveraging AWS Bedrock and open/internal LLMs with function calling and RAG, enabling conversational natural-language querying of high-dimensional datasets.
- Directed the design of a multi-model fraud detection architecture for a Tier-1 US credit bureau spanning first-party, third-party, and synthetic identity risk, integrating behavioral biometrics signals into real-time decisioning logic.
- Served as primary technical lead and code owner for Python, PySpark, and AWS workflows leading a team of 6–8 onshore/offshore data scientists and engineers, establishing repository standards, CI/CD reviews, and mentorship.
- Led performance tuning and query optimization for large-scale Spark/SQL jobs on AWS EMR, achieving an 80% speedup in processing speed and 60% savings in AWS compute spend.
- Partnered with enterprise client leaders to translate high-dimensional machine learning outputs into core commercial value, designing custom predictive solutions that drove enterprise software adoption.
- Deployed end-to-end production data science pipelines orchestrated via Apache Airflow and AWS, maintaining low-latency inference pipelines and high-throughput batch scoring jobs.

ASSOCIATE LEAD DATA SCIENTIST

- Led Ad-Tech analytics and causal inference initiatives for a Fortune 500 global consumer goods (CPG) company's marketing attribution system, redesigning the sales contribution decomposition model and presenting insights across multiple global brands to optimize system performance.
- Developed a proprietary audience segmentation algorithm to cluster website traffic using Leiden's method, enabling targeted campaigns that drove a 40% increase in engagement rate.
- Built ML-driven system optimization for sponsored ads spend allocation on Amazon, improving product ROAS by 15%.
- Engineered an identity graph system for a consumer-intent data platform (Ad-Tech) using ML algorithms on distributed big data platforms to classify identifiers, achieving a ~10% reduction in data wastage.
- Built an XGBoost and elastic-net model ensemble for a leading Latin American beauty & cosmetics enterprise to do demand forecasting, improving supply chain system efficiency, reducing error rate by 52%, and reducing annual operating cost by \$3M through predictive supply chain optimization.

CYIENT INSIGHTS, Hyderabad

Jul 2016 - Jul 2018

DATA SCIENTIST

- Developed an analytics solution for a leading mining-equipment wear-parts manufacturer to detect loss of excavator tooth in the field within 60 seconds using sensor data, enhancing product safety and avoiding downtime cost of ~\$1.5M.
- Built predictive models for a leading aerospace engine manufacturer to predict unplanned maintenance based on usage parameters, operator history, and environment details, achieving an R^2 of 0.87 and enabling customized maintenance plans for the OEM.
- Applied artificial neural networks to satellite imagery, supporting a global navigation & mapping technology company for product development in remote mapping and annotation.
- Created Power BI dashboards for a global rail OEM to help analyze live warning signals by priority and correlation.

PROJECTS & TECHNICAL LABS

- **Agentic Script-to-Visual Generation Pipeline** (*Python, PyTorch, Diffusers, Gemini API*): Built an end-to-end multimodal workflow converting raw scripts into structured screenplays, shot-by-shot prompt breakdowns, and rendered visual storyboards. Orchestrated LLM agents for text decomposition and deployed open-source diffusion models locally with optimized VRAM usage and model quantization.
- **Voice Recognition & Speaker Identification** (*PyTorch, CNNs, Siamese Networks*): Built a one-shot learning voice recognition system using a 15-layer CNN on VoxCeleb audio spectrograms, applying Siamese networks and triplet-loss with cosine distance to achieve an AUC of 0.94 on unseen speakers.
- **Rapid Mobile Application Prototyping** (*Android, Kotlin, AI-Assisted Dev*): Designed and published two Android applications for poker strategy modeling and personal interest tracking, using LLM-assisted code generation and rapid prototyping to deliver functional mobile UIs and embedded logic within tight development cycles.

INTERNSHIPS

Indian Space Research Organisation (ISRO), Bengaluru

May 2015 - Jun 2015

- Worked on inverse GPS-like architecture, contributing to product accuracy improvements in satellite tracking.

Airports Authority of India (AAI), Bengaluru

May 2014 - Jun 2014

- Developed a MATLAB user interface for GAGAN, enhancing product reliability in navigation through image processing.

AWARDS & CERTIFICATIONS

- Generative AI & LLM Architectures - Deep Learning & Transformer Architectures (Self-Directed Study)
- Stanford Online - Machine Learning, Andrew Ng
- DeepLearning.AI - Deep Learning Specialization (5-course), Coursera
- SAS Certified Base Programmer for SAS 9
- Received multiple Tiger Analytics Spot Awards for outstanding technical leadership and high-impact client delivery
- Received Value Award for Ownership & Accountability at Sigmoid Analytics, 2019